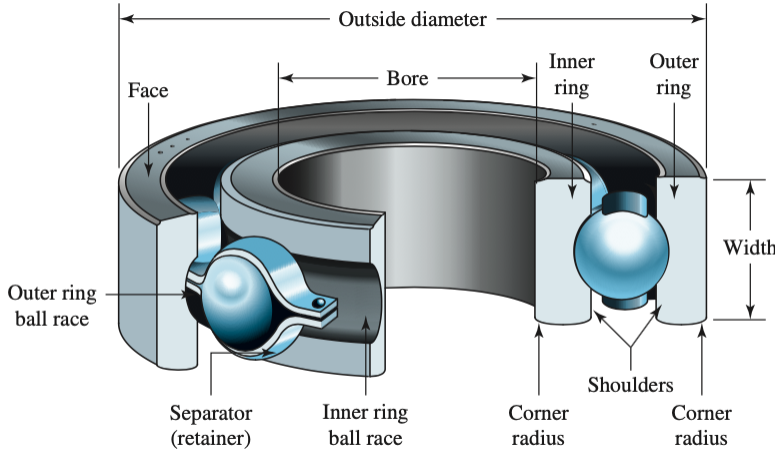


4.2 Ball and Cylindrical Roller Bearings

Types of bearings that fall under this category are:

- Deep-groove ball bearings
- Angular deep-groove ball bearings
- Cylindrical roller bearings

4.2.1 Anatomy



4.2.2 Design Selection - Radial Load Only

1. Find the angular speed n_D (in rpm) if it is not already given to you.
 - (a) Find the transmitted torque from a free-body-diagram using $T = Fd$. It will be the torque that causes the shaft to rotate along its axis.
 - (b) If the design horsepower is given, use the following equation:

$$n_D = \frac{63025H}{T} \text{ where } H \text{ is the design horsepower in hp,}$$
$$T \text{ is the torque transmitted in lbf} \cdot \text{in}$$

2. Find the radial load by taking the magnitude of the radial force components:

$$F_R = \sqrt{F_x^2 + F_y^2}$$

3. Calculate the bearing reliability of each individual bearing (if the total reliability of the ensemble is given):

$$R_i = \sqrt[n]{R_{tot}}$$

where R_i is the individual bearing reliability,

R_{tot} is the total bearing reliability,

n is the number of bearings in the assembly

4. Get bearing design life $\mathcal{L}_D$ for your application from this table (if not given in the question):

Table 11–4 Bearing-Life Recommendations for Various Classes of Machinery

Type of Application	Life, kh
Instruments and apparatus for infrequent use	Up to 0.5
Aircraft engines	0.5–2
Machines for short or intermittent operation where service interruption is of minor importance	4–8
Machines for intermittent service where reliable operation is of great importance	8–14
Machines for 8-h service that are not always fully utilized	14–20
Machines for 8-h service that are fully utilized	20–30
Machines for continuous 24-h service	50–60
Machines for continuous 24-h service where reliability is of extreme importance	100–200

5. Calculate the multiple of rating life for each bearing in the assembly:

$$x_D = \frac{L_D}{L_{10}} = \frac{60\mathcal{L}_D n_D}{L_{10}}$$

where L_D is bearing design life in number of revolutions,

L_{10} is the rating life,

$\mathcal{L}_D$ is the design life in hours,

n_D is the angular speed of the bearing in rpm

The Weibull parameters used will depend on which manufacturer's bearings we are using.

- Timken (Manufacturer 1) is common for tapered roller bearings
- SKF (Manufacturer 2) is common for ball and straight roller bearings

Manufacturer	Rating Life, Revolutions	Weibull Parameters Rating Lives		
		X_θ	θ	b
1	90(10 ⁶)	0	4.48	1.5
2	1(10 ⁶)	0.02	4.459	1.483

6. Get application factor from this table (if not given in the question):

Table 11–5 Load-Application Factors

Type of Application	Load Factor
Precision gearing	1.0–1.1
Commercial gearing	1.1–1.3
Applications with poor bearing seals	1.2
Machinery with no impact	1.0–1.2
Machinery with light impact	1.2–1.5
Machinery with moderate impact	1.5–3.0

7. Calculate the load rating C_{10} for the radial load:

$$C_{10} = a_f F_D \left[\frac{x_D}{x_0 + (\theta - x_0) [\ln(1/R_D)]^{1/b}} \right]^{1/a}$$

- $a = 3$ for ball bearings
- $a = 10/3$ for roller bearings (cylindrical and tapered)

8. Select a bearing from one of these tables that has a C_{10} value greater than the calculated one.

- For deep-groove and angular-contact ball bearings:

Table 11–2 Dimensions and Load Ratings for Single-Row 02-Series Deep-Groove and Angular-Contact Ball Bearings

Bore, mm	OD, mm	Width, mm	Fillet Radius, mm	Shoulder		Load Ratings, kN			
				Diameter, mm		Deep Groove		Angular Contact	
				d_s	d_H	C_{10}	C_0	C_{10}	C_0
10	30	9	0.6	12.5	27	5.07	2.24	4.94	2.12
12	32	10	0.6	14.5	28	6.89	3.10	7.02	3.05
15	35	11	0.6	17.5	31	7.80	3.55	8.06	3.65
17	40	12	0.6	19.5	34	9.56	4.50	9.95	4.75
20	47	14	1.0	25	41	12.7	6.20	13.3	6.55
25	52	15	1.0	30	47	14.0	6.95	14.8	7.65
30	62	16	1.0	35	55	19.5	10.0	20.3	11.0
35	72	17	1.0	41	65	25.5	13.7	27.0	15.0
40	80	18	1.0	46	72	30.7	16.6	31.9	18.6
45	85	19	1.0	52	77	33.2	18.6	35.8	21.2
50	90	20	1.0	56	82	35.1	19.6	37.7	22.8
55	100	21	1.5	63	90	43.6	25.0	46.2	28.5
60	110	22	1.5	70	99	47.5	28.0	55.9	35.5
65	120	23	1.5	74	109	55.9	34.0	63.7	41.5
70	125	24	1.5	79	114	61.8	37.5	68.9	45.5
75	130	25	1.5	86	119	66.3	40.5	71.5	49.0
80	140	26	2.0	93	127	70.2	45.0	80.6	55.0
85	150	28	2.0	99	136	83.2	53.0	90.4	63.0
90	160	30	2.0	104	146	95.6	62.0	106	73.5
95	170	32	2.0	110	156	108	69.5	121	85.0

- For cylindrical roller bearings:

Table 11–3 Dimensions and Basic Load Ratings for Cylindrical Roller Bearings

	02-Series				03-Series			
Bore, mm	OD, mm	Width, mm	Load Rating, kN		OD, mm	Width, mm	Load Rating, kN	
			C_{10}	C_0			C_{10}	C_0
25	52	15	16.8	8.8	62	17	28.6	15.0
30	62	16	22.4	12.0	72	19	36.9	20.0
35	72	17	31.9	17.6	80	21	44.6	27.1
40	80	18	41.8	24.0	90	23	56.1	32.5
45	85	19	44.0	25.5	100	25	72.1	45.4
50	90	20	45.7	27.5	110	27	88.0	52.0
55	100	21	56.1	34.0	120	29	102	67.2
60	110	22	64.4	43.1	130	31	123	76.5
65	120	23	76.5	51.2	140	33	138	85.0
70	125	24	79.2	51.2	150	35	151	102
75	130	25	93.1	63.2	160	37	183	125
80	140	26	106	69.4	170	39	190	125
85	150	28	119	78.3	180	41	212	149
90	160	30	142	100	190	43	242	160
95	170	32	165	112	200	45	264	189
100	180	34	183	125	215	47	303	220
110	200	38	229	167	240	50	391	304
120	215	40	260	183	260	55	457	340
130	230	40	270	193	280	58	539	408
140	250	42	319	240	300	62	682	454
150	270	45	446	260	320	65	781	502

9. We're done!

4.2.3 Design Selection - Radial and Thrust Loads

1. Determine the rotation factor V :

- $V = 1$ when the inner ring rotates
- $V = 1.2$ when the outer ring rotates

2. If we do not yet have the C_0 and C_{10} value for the bearing, we will have to make assumptions:

- Assume $F_a/(VF_r) > e$ (we will check if this is actually true later)
- Choose random values for the X_2 and Y_2 factors. It's best to just choose middle values on the table, so let's go with $X_2 = 0.56$ and $Y_2 = 1.63$.

Table 11–1 Equivalent Radial Load Factors for Ball Bearings

F_a/C_0	e	$F_a/(VF_r) \leq e$		$F_a/(VF_r) > e$	
		X_1	Y_1	X_2	Y_2
0.014*	0.19	1.00	0	0.56	2.30
0.021	0.21	1.00	0	0.56	2.15
0.028	0.22	1.00	0	0.56	1.99
0.042	0.24	1.00	0	0.56	1.85
0.056	0.26	1.00	0	0.56	1.71
0.070	0.27	1.00	0	0.56	1.63
0.084	0.28	1.00	0	0.56	1.55
0.110	0.30	1.00	0	0.56	1.45
0.17	0.34	1.00	0	0.56	1.31
0.28	0.38	1.00	0	0.56	1.15
0.42	0.42	1.00	0	0.56	1.04
0.56	0.44	1.00	0	0.56	1.00

*Use 0.014 if $F_a/C_0 < 0.014$.

- Using our assumptions, calculate the equivalent load using the following equation:

$$F_e = X_i V F_r + Y_i F_a$$

where F_r is the radial load,

F_a is the axial/thrust load

- Calculate the load rating C_{10} for the equivalent load:

$$C_{10} = a_f F_e \left[\frac{x_D}{x_0 + (\theta - x_0) [\ln(1/R_D)]^{1/b}} \right]^{1/a}$$

- $a = 3$ for ball bearings
 - $a = 10/3$ for roller bearings (cylindrical and tapered)
- Select a bearing that has a C_{10} value that exceeds the calculated one. Refer to tables 11-12 and 11-13 in **step 8** of the radial section right above.
 - Find the C_0 value for the chosen bearing.
 - Calculate F_a/C_0 where F_a is the axial load.
 - Look back at table 11-1 in **step 2** of this section and find the closest value of F_a/C_0 . Get e from this.
 - Check if $F_a/(VF_r) > e$. If so, we will be looking for Y_2 values in table 11-1 from **step 2**. If not, we will be looking for Y_1 values.

10. Interpolate to find Y_2 .

- The two closest F_a/C_0 values in the table will be the x-points. The two associated Y_2 values will be the y-points.
- Calculate the slope: $m = \frac{y_1 - y_2}{x_1 - x_2}$
- Plug in a point into $y = mx + b$ and solve for b
- Finally, plug in your value of F_a/C_0 as x to get Y_2 as y

11. Recalculate F_e with the new Y_2 value:

$$F_e = X_i V F_r + Y_i F_a$$

12. Calculate the new C_{10} . The calculation for C_{10} only changes in F_e so we can do the following calculation:

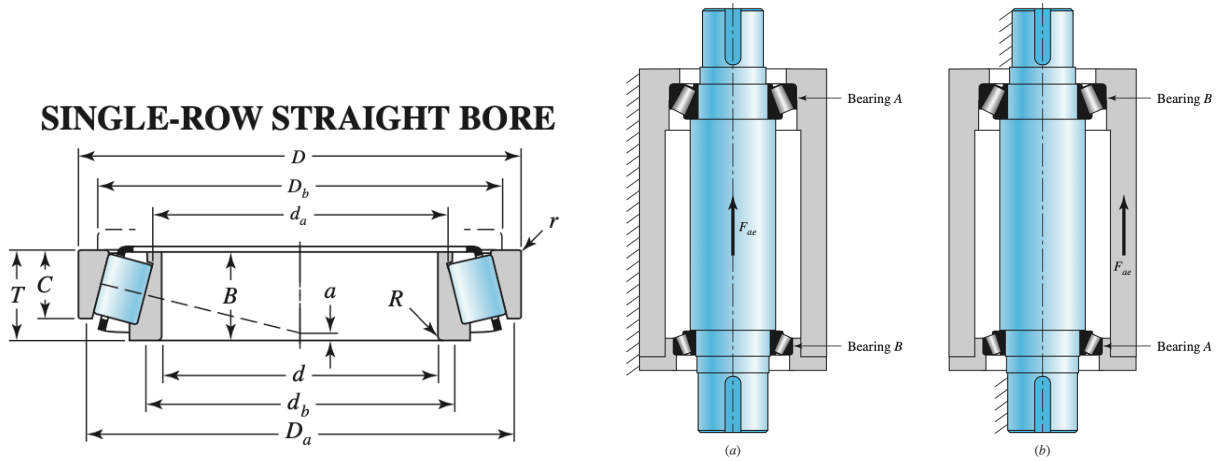
$$\text{new } C_{10} = \frac{\text{new } F_e}{\text{old } F_e} (\text{old } C_{10})$$

13. Select a bearing that has a C_{10} value that exceeds the calculated one. Refer to tables 11-2 or 11-3 from **step 8** of the previous section.

14. Repeat **steps 6 to 13** until you select the same bearing from the table twice in a row.

4.3 Tapered Roller Bearings

4.3.1 Anatomy



4.3.2 Design Selection

1. Calculate the radial and axial loads on each bearing

- $F_{rA} = \sqrt{R_{xA}^2 + R_{yA}^2}$
- $F_{rB} = \sqrt{R_{xB}^2 + R_{yB}^2}$